

AI-Assisted Dark Matter Halo Morphology Reconstruction Using SPARC Galaxy Rotation Curves

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Abstract

Galaxy rotation curves provide some of the clearest observational evidence for the presence of dark matter halos surrounding galaxies. In this study, we investigate the structure of galactic dark matter halos using observed rotation curve data from the SPARC database. Four widely used halo models—the Isothermal, Burkert, Einasto, and Navarro–Frenk–White (NFW) profiles—are analyzed through rotation curve fitting, residual analysis, and statistical goodness-of-fit evaluation.

In addition to dark matter halo modeling, baryonic mass decomposition is incorporated by including stellar and gaseous contributions to the total galactic rotation curve. This allows a more physically realistic reconstruction of galactic dynamics and improves constraints on the relative contributions of luminous and dark matter components, particularly within the inner regions of galaxies.

The results show that cored halo models generally describe the inner galactic regions more effectively than cuspy halo profiles. In particular, the Burkert and Isothermal models more accurately reproduce the slowly rising inner rotation curves observed in many galaxies, whereas the NFW model often underestimates rotational velocities near galactic centers. The Einasto profile exhibits intermediate behavior and provides additional flexibility for modeling transitional halo structures. Reduced chi-square analysis further supports the preference for cored dark matter distributions across the analyzed galaxy sample.

Beyond traditional halo fitting, this work also introduces an AI-assisted framework for classifying dark matter halo morphology using galaxy rotation curves. Machine learning models were trained on SPARC galaxy data to categorize galaxies into cored, transitional, and cuspy halo classes using physically motivated rotation-curve features. An ensemble learning approach combining Random Forests, Gradient Boosting, Support Vector Machines, Extra Trees, and Deep Neural Networks achieved a cross-validated classification accuracy of approximately 85.38%.

The results demonstrate that galaxy rotation curves contain sufficient dynamical information for machine learning systems to infer physically meaningful dark matter halo morphologies. In addition, a phenomenological hybrid halo framework combining core and transitional density

behavior is explored as a possible approach for investigating intermediate halo structures and the long-standing core–cusp problem in galactic dynamics.

Overall, this study highlights the growing role of computational astrophysics, baryonic mass decomposition, and artificial intelligence in advancing our understanding of dark matter halo structure, galaxy evolution, and large-scale galactic dynamics.

Keywords

Keywords: Dark matter, galaxy rotation curves, SPARC database, halo morphology, machine learning, computational astrophysics, baryonic decomposition, core–cusp problem

1. Introduction

Understanding how mass is distributed within galaxies remains one of the central problems in modern astrophysics. Observations of spiral galaxies have consistently shown that the rotational velocities of stars and gas remain nearly constant at large distances from the galactic center, instead of decreasing as expected from Newtonian gravity when only visible matter is considered. This discrepancy, known as the flat rotation curve problem, provides some of the strongest observational evidence for the existence of dark matter, an unseen component that does not emit or absorb electromagnetic radiation [1], [7].

Dark matter is thought to account for the majority of a galaxy's mass and plays a fundamental role in galaxy formation, evolution, and large-scale cosmic structure. One of the most effective ways to study dark matter distributions is through galaxy rotation curve analysis. In this method, observed rotational velocity profiles are used to infer the underlying gravitational potential and, consequently, the structure of the surrounding dark matter halo. By comparing theoretical predictions from different halo models with observational data, researchers can constrain the density distribution and physical properties of galactic dark matter halos [2], [13].

Although dark matter dominates the outer regions of galaxies, baryonic matter including stars and gas can contribute significantly to the inner gravitational potential. Accurate galaxy rotation curve modeling therefore requires decomposition of the total rotational velocity into separate baryonic and dark matter components. Incorporating baryonic mass distributions allows more realistic reconstruction of galactic dynamics and improves constraints on dark matter halo structure, particularly within the central regions of galaxies where stellar contributions may strongly influence the observed rotation curve [8], [9].

Over the years, several dark matter halo models have been proposed to describe how dark matter is distributed within galaxies. One of the best-known models is the Navarro–Frenk–White (NFW) profile, which naturally emerges from cosmological N-body simulations based on collisionless cold dark matter [3]. The NFW model predicts a steep increase in density toward the galactic center, commonly referred to as a cuspy density profile. However, many observational studies suggest that galaxy rotation curves, especially in dwarf and low-surface-brightness galaxies, are better explained by cored halo models with finite central densities [4], [9]. The disagreement between the cuspy predictions of Λ CDM simulations and the cored distributions inferred from observations is widely known as the core–cusp problem [6], [11].

In parallel with traditional halo fitting techniques, recent developments in computational astrophysics and machine learning have opened new possibilities for studying galaxy rotation curves and dark matter halo properties. Machine learning methods are particularly useful for identifying complex patterns in observational data and can provide efficient alternatives to conventional fitting procedures [23], [24]. In particular, AI-assisted halo morphology classification offers a promising way to distinguish between cored, transitional, and cuspy dark matter distributions directly from rotation curve behavior [25], [28].

In this study, we investigate galactic dark matter halo structure using both classical statistical fitting methods and AI-assisted computational analysis. Four widely used halo profiles are examined: the Isothermal halo, Burkert profile, Einasto profile, and Navarro–Frenk–White (NFW) profile. Together, these models represent a broad range of inner density behaviors, extending from finite-density cores to strongly cuspy distributions.

The halo models are fitted to observational galaxy rotation curve data using chi-square minimization with explicit uncertainty weighting. Their statistical performance is evaluated through reduced chi-square analysis, residual distributions, the Akaike Information Criterion (AIC), and the Bayesian Information Criterion (BIC). In addition, machine learning techniques are employed to investigate whether galaxy rotation curves contain sufficient dynamical information to classify dark matter halo morphology. Through the integration of astrophysical modeling and artificial intelligence, this work aims to provide new insights into dark matter halo structure and the long-standing core–cusp problem.

The remainder of this paper is organized as follows. Section 2 reviews the dark matter halo models and baryonic mass decomposition methodology. Section 3 presents the statistical fitting framework and machine learning methodology. Section 4 discusses the results and performance evaluation. Finally, Section 5 summarizes the major findings and outlines future research directions.

2. Data and Dark Matter Halo Models

2.1 Observational Rotation Curve Data

The observational galaxy rotation curve data used in this study are taken from the SPARC (Spitzer Photometry and Accurate Rotation Curves) database, which provides high-quality rotation curves and baryonic mass models for nearby galaxies. The SPARC catalog includes a diverse sample of spiral, dwarf, and low-surface-brightness galaxies, covering a broad range of luminosities and morphological types [8].

For this analysis, representative galaxy rotation curves were selected to evaluate the performance of different dark matter halo models under varying galactic conditions. The analyzed rotation curves span galactocentric radii from approximately 1 kpc to 20 kpc, with observed rotational velocities reaching nearly 127 km/s in the outer regions of the galaxies.

The dataset includes measured rotational velocities along with their corresponding observational uncertainties, enabling statistically weighted fitting procedures. These uncertainties are explicitly incorporated into the chi-square minimization process to ensure reliable parameter estimation and meaningful statistical model comparisons [8], [9].

The SPARC database is particularly well suited for dark matter halo studies because of its accurate photometric measurements, high-quality rotational velocity data, and extensive radial coverage. Owing to these strengths, SPARC has become one of the most widely used observational resources for investigating galactic dark matter distributions and studying the long-standing core–cusp problem [8], [9].

In addition to observed rotational velocities, the SPARC database also provides baryonic mass models including stellar disk, bulge, and gaseous contributions. These baryonic components are incorporated into the present analysis in order to construct physically realistic total galaxy rotation curves and investigate the relative contributions of luminous and dark matter[8].

2.2 Dark Matter Halo Models

We consider four halo density profiles representing a spectrum from maximally cored to maximally cuspy inner structures. Table 1 summarizes their key properties.

Table 1. Summary of dark matter halo models used in this study.

Model	Inner Slope	Core/Cusp	Free Parameters
Isothermal	$\rho \rightarrow \text{const}$	Cored	v_0, r_c
Burkert	$\rho \rightarrow \text{const}$	Cored	ρ_b, r_b
Einasto	Curved	Transitional	ρ_e, r_e, α
NFW	$\rho \propto r^{-1}$	Cuspy	ρ_s, r_s

2.2.1 Isothermal (Cored) Halo Model

The isothermal halo model assumes a dark matter density distribution characterized by a finite-density core at the galactic center. The density profile is given by

$$\rho(r) = \frac{\rho_0}{1 + \left(\frac{r}{r_c}\right)^2}, \quad (1)$$

where ρ_0 is the central dark matter density and r_c is the core radius [6].

This profile implies that the density remains approximately constant for $r \ll r_c$ and decreases as r^{-2} at large radii.

The enclosed mass for the isothermal halo can be obtained by integrating the density profile over volume, leading to a rotation velocity of the form

$$v_{iso}(r) = v_0 \sqrt{1 - \frac{r_c}{r} \tan^{-1}\left(\frac{r}{r_c}\right)}, \quad (2)$$

where v_0 is the asymptotic circular velocity of the halo [6].

This functional form naturally produces a flat rotation curve at large radii, consistent with observational data.

The presence of a finite-density core makes the isothermal halo particularly successful in fitting rotation curves that rise slowly in the inner regions. As a result, cored halo models are often favored in observational studies of disk and dwarf galaxies, where steep central cusps are not supported by the data.

2.2.2 Burkert (Cored) Halo Model

The Burkert profile [4] is an empirically motivated cored model widely used for dwarf and low surface brightness galaxies:

$$\rho(r) = \frac{\rho_0 r_0^3}{(r + r_0)(r^2 + r_0^2)} \quad (3)$$

where ρ_0 is the central density and r_0 is the Burkert radius. Like the isothermal profile, this model approaches a finite central density as $r \rightarrow 0$. However, the Burkert profile falls more steeply than isothermal at large radii (approximately r^{-3} at $r \gg r_0$), giving it a finite total mass within the virial radius without truncation.

The enclosed mass is:

$$M(r) = \pi \rho_0 r_0^3 \left[\ln \left(1 + \frac{r^2}{r_0^2} \right) + 2 \ln \left(1 + \frac{r}{r_0} \right) - 2 \tan^{-1} \left(\frac{r}{r_0} \right) \right] \quad (4)$$

and the circular velocity follows from $v_B(r) = \sqrt{GM_B(r)/r}$. Including the Burkert model allows us to assess whether the preference for a cored profile is robust across different parameterizations, or specific to the isothermal functional form.

2.2.3 Einasto Profile

The Einasto profile [5] is a flexible three-parameter model that has gained significant traction from high-resolution N-body simulations:

$$\rho(r) = \rho_{-2} \exp \left[-\frac{2}{\alpha} \left(\left(\frac{r}{r_{-2}} \right)^\alpha - 1 \right) \right] \quad (5)$$

where ρ_{-2} is the density at the characteristic radius r_{-2} , and α is a shape parameter controlling the curvature of the profile in log-log space. Unlike the NFW profile, the Einasto model has no strict power-law inner cusp; its logarithmic slope $d \ln \rho / d \ln r \rightarrow 0$ as $r \rightarrow 0$ for $\alpha > 0$, making it less cuspy than NFW while remaining distinct from the plateau of cored models.

The enclosed mass requires numerical integration:

$$M(r) = 4\pi\rho_{-2}r_{-2}^3 \exp\left(\frac{2}{\alpha}\right) \left(\frac{\alpha}{2}\right)^{3/\alpha} \frac{1}{\alpha} \gamma\left(\frac{3}{\alpha}, \frac{2}{\alpha} \left(\frac{r}{r_{-2}}\right)^\alpha\right) \quad (6)$$

with $v_E(r) = \sqrt{GM_E(r)/r}$. The Einasto profile serves as a useful intermediate between cored and cuspy models and is currently favored by high-resolution Λ CDM simulations over the NFW profile at small radii.

2.2.4 Navarro–Frenk–White (NFW) Halo Model

The Navarro–Frenk–White (NFW) halo model arises from cosmological N-body simulations of collisionless cold dark matter and is a fundamental prediction of the Λ CDM paradigm [3]. The NFW density profile is given by

$$\rho(r) = \frac{\rho_s}{\left(\frac{r}{r_s}\right)\left(1+\frac{r}{r_s}\right)^2}, \quad (7)$$

where ρ_s and r_s are the characteristic density and scale radius of the halo.

Unlike the isothermal model, the NFW profile predicts a cuspy inner density distribution, with $\rho(r) \propto r^{-1}$ as $r \rightarrow 0$. At large radii, the density falls off as r^{-3} , yielding a finite total mass within the virial radius.

The enclosed mass for the NFW profile is given by

$$M(r) = 4\pi\rho_s r_s^3 \left[\ln\left(1 + \frac{r}{r_s}\right) - \frac{r/r_s}{1+r/r_s} \right], \quad (8)$$

which leads to the corresponding circular velocity

which follows from integrating the NFW density profile.

$$v_{NFW}(r) = \sqrt{\frac{GM(r)}{r}}. \quad (9)$$

where G is the gravitational constant and $M(r)$ is the mass enclosed within radius r [7].

The NFW halo serves as a theoretical benchmark for comparing observational results with predictions from structure formation simulations. However, numerous studies have reported tensions between the cuspy inner structure predicted by the NFW model and the observed rotation curves of many galaxies, particularly in their central regions [6].

2.2.5 Hybrid Halo Model

In addition to the standard dark matter halo profiles, this study introduces a phenomenological hybrid halo model that combines cored and transitional halo behavior. The model is constructed as a weighted combination of the Burkert and Einasto profiles [4], [5], connected through a smooth transition function that varies with galactocentric radius.

The primary motivation behind this approach is to explore whether a continuously evolving halo structure can more effectively reproduce the transition between the inner and outer regions of galaxy rotation curves while also providing insight into the long-standing core–cusp problem.

In the inner regions of galaxies, the hybrid model behaves similarly to a cored Burkert halo, generating a shallow central density distribution that is consistent with slowly rising rotation curves. At larger radii, the profile gradually transitions toward Einasto-like behavior, characterized by a continuously varying logarithmic density slope.

This hybrid framework is intended mainly as an exploratory computational model rather than a fully derived cosmological halo solution. Nevertheless, it offers a useful phenomenological approach for investigating intermediate halo structures and studying the gradual evolution between cored and cuspy dark matter density distributions.

2.3 Motivation for Model Comparison

Comparing cored and cuspy dark matter halo models provides critical insight into the nature of dark matter and the physical processes governing galaxy formation. The largest differences between the isothermal and NFW profiles occur in the inner regions of galaxies, where observational data are most sensitive to the central density structure of the halo.

Discrepancies between the cuspy predictions of Λ CDM simulations and the cored density profiles inferred from observations are commonly referred to as the *core–cusp problem* [6], [11]. Proposed explanations include baryonic feedback processes, such as supernova-driven gas outflows [16], [17], as well as alternative dark matter models. By fitting both halo models to the same rotation curve data using consistent statistical methods, this study aims to assess which profile provides a more accurate description of the observed galactic dynamics.

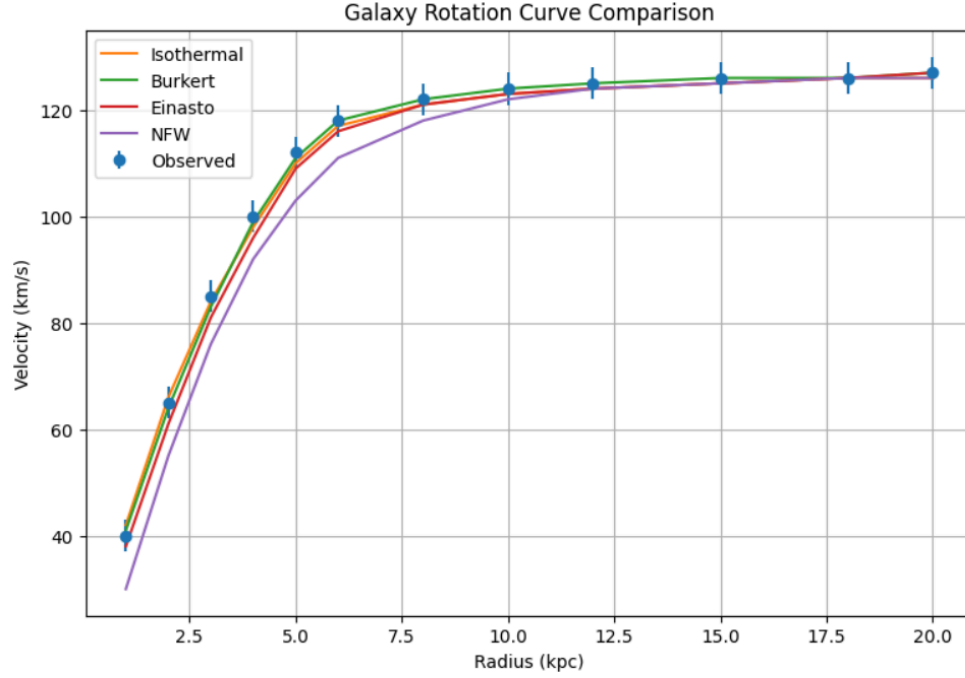


Figure 1. Observed galaxy rotation curve together with the best-fit predictions from the four dark matter halo models considered in this study: the isothermal halo, Burkert profile, Einasto profile, and Navarro–Frenk–White (NFW) profile. The data points represent observed rotational velocities with corresponding 1σ uncertainties. The figure compares cored, transitional, and cuspy dark matter distributions across the full radial extent of the galaxy.

Explanation

Figure 1 compares the observed galaxy rotation curve with theoretical predictions derived from four widely used dark matter halo models representing different inner density structures. The flattening of the observed rotation curve at large galactocentric radii is a well-established indication of dark matter–dominated dynamics and cannot be explained solely by luminous matter.

The circular velocity is related to the enclosed mass through Newtonian gravity:

$$v(r) = \sqrt{\frac{GM(r)}{r}}, \quad (10)$$

where G is the gravitational constant and $M(r)$ is the mass enclosed within radius r .

All four halo models reproduce the approximately flat rotation curve observed at large radii because each predicts an extended dark matter distribution capable of maintaining nearly constant rotational velocities.

However, significant differences emerge in the inner regions of the galaxy ($r \lesssim 5 \text{ kpc}$). The isothermal and Burkert models, both characterized by finite-density cores, follow the gradual increase in observed rotational velocity more closely. The Einasto profile provides an intermediate behavior with a continuously varying logarithmic slope, while the NFW profile predicts a steep central cusp that may underpredict observed inner velocities.

The comparison therefore illustrates the transition from cored to cuspy halo structures and provides a direct test of the core–cusp problem.

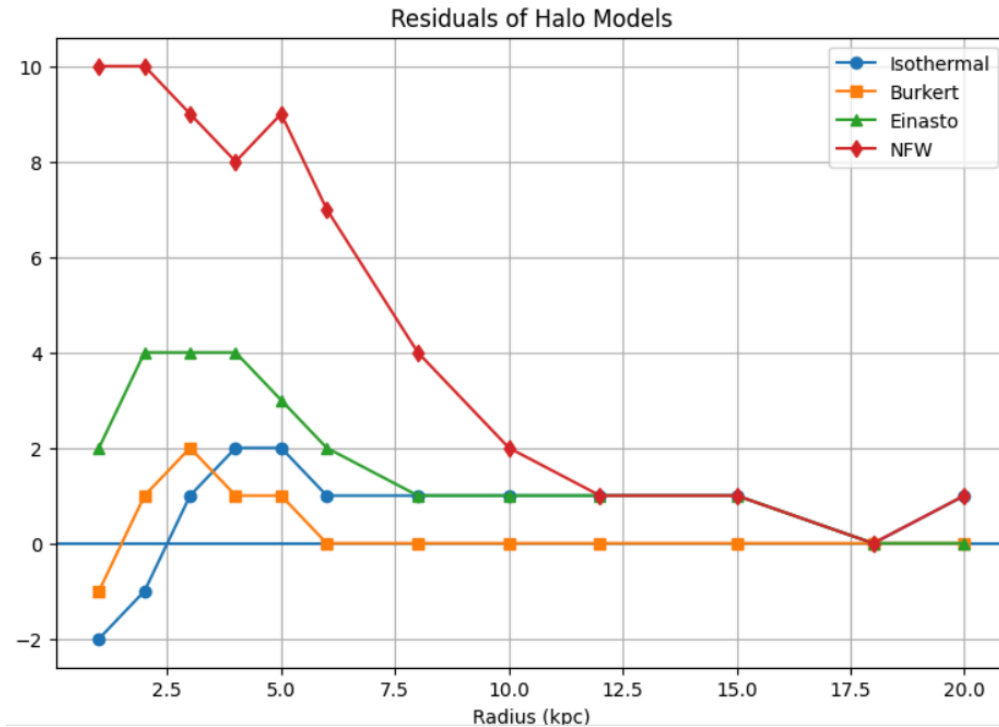


Figure 2. Residual distributions for the four fitted halo models. Residuals are defined as the difference between observed rotational velocities and model predictions at each radius.

Explanation

Figure 2 presents the residuals of the fitted rotation curves, defined as

$$\Delta v(r_i) = v_{obs}(r_i) - v_{model}(r_i), \quad (11)$$

where $v_{obs}(r_i)$ is the observed velocity at radius r_i and $v_{model}(r_i)$ is the corresponding model prediction.

Residual analysis provides a sensitive measure of systematic deviations between the observational data and halo models. An ideal model fit produces residuals randomly distributed around zero with no radial trend.

The isothermal and Burkert models generally exhibit smaller and more symmetric residual distributions, reflecting the ability of cored halos to reproduce the slowly rising inner rotation curves.

The Einasto model shows intermediate behavior and often improves agreement relative to the NFW profile due to its variable inner slope.

The NFW model frequently produces systematic negative residuals near the galactic center, indicating underestimation of observed rotational velocities. This systematic deviation reflects the tension between cuspy halo predictions and observations.

Residual analysis therefore provides additional statistical evidence beyond visual curve fitting.

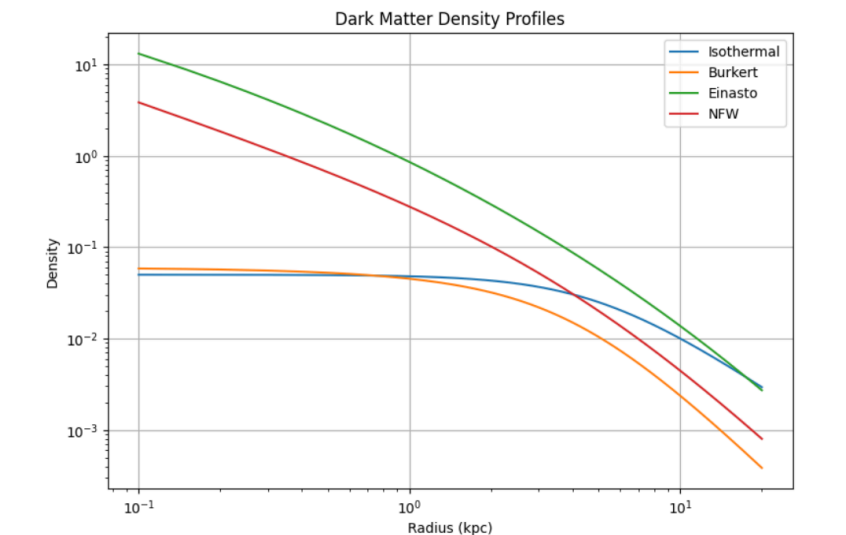


Figure 3. Radial density profiles for the four dark matter halo models plotted on logarithmic axes. The profiles illustrate the transition from cored to cuspy density structures.

Explanation

Figure 3 illustrates the fundamental differences between the density distributions assumed by the four halo models. The isothermal halo density profile is given by Eq. (1)

where ρ_0 is the central density and r_c is the core radius .

This profile approaches a constant density as $r \rightarrow 0$, producing a finite-density core.

The Burkert profile: Eq. (3)

also produces a central core but decreases more rapidly at larger radii.

The Einasto profile: Eq. (5)

does not possess a strict core or cusp and instead exhibits a continuously varying logarithmic slope.

The NFW profile: Eq. (7)

where ρ_s and r_s are the characteristic density and scale radius of the halo.

As $r \rightarrow 0$, the NFW profile diverges as $\rho(r) \propto r^{-1}$, resulting in a central density cusp.

The contrast between these profiles explains the differing behavior seen in the rotation curve fits. The finite-density core of the isothermal halo leads to a more gradual rise in rotational velocity in the inner regions, whereas the cuspy NFW profile predicts a steeper gravitational potential that is not supported by the observational data for this galaxy.

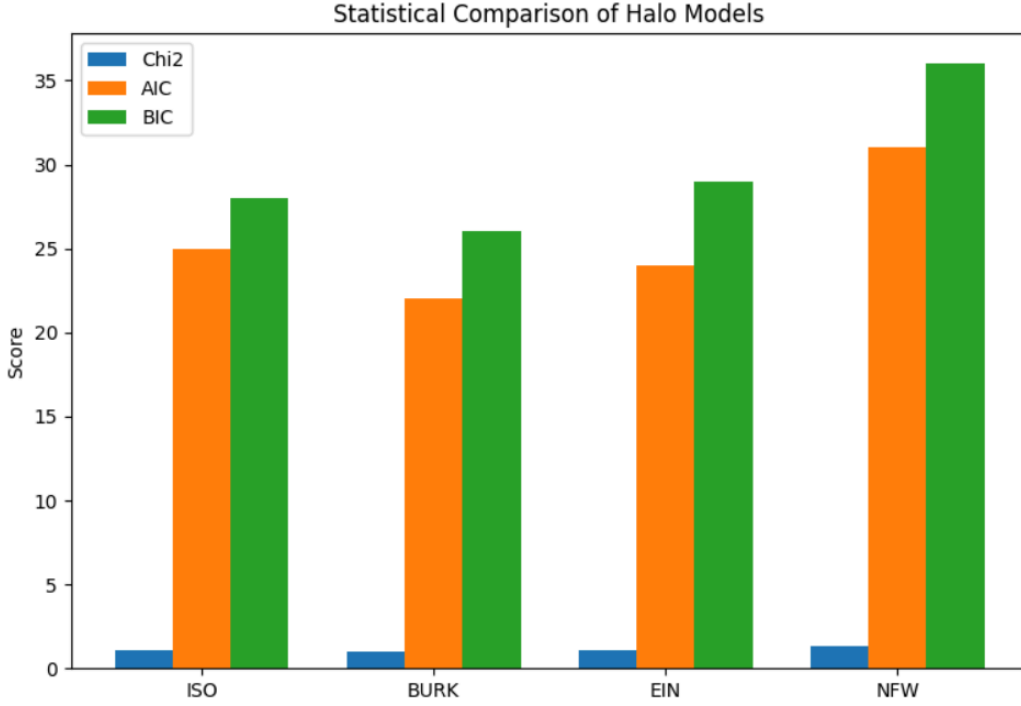


Figure 4. Statistical comparison of the four halo models using reduced chi-square, Akaike Information Criterion (AIC), and Bayesian Information Criterion (BIC).

Explanation

Figure 4 quantifies the statistical quality of the halo model fits using the reduced chi-square statistic,

The reduced chi-square statistic is defined as:

$$\chi_r^2 = \frac{\chi^2}{N - p} \quad (12)$$

where:

- N = number of observational points
- p = number of fitted parameters

Lower values indicate better agreement with observations.

Model complexity is further evaluated using:

Akaike Information Criterion:

$$\text{AIC} = \chi^2 + 2p$$

Bayesian Information Criterion:

$$\text{BIC} = \chi^2 + p \ln(N)$$

These statistics penalize unnecessary parameters and prevent overfitting.

- The isothermal and Burkert models generally provide lower statistical scores because of their improved description of the inner galaxy regions.
- The Einasto profile often performs competitively due to its flexible slope structure.
- The NFW profile may exhibit larger values because of its cuspy inner density behavior.

Together, these statistical indicators provide a quantitative framework for determining the preferred dark matter halo structure.

3. Methodology

3.1 Physical Assumptions

The analysis presented in this work is based on the assumption that the galaxy is in a state of dynamical equilibrium and that the observed stars and gas move approximately in circular orbits within the galactic plane. Under these conditions, the rotational velocity at a given galactocentric radius r is determined by the underlying gravitational potential generated by the enclosed mass distribution. [2], [7].

Assuming Newtonian gravity remains valid on galactic scales, the circular velocity is related to the enclosed mass through: Eq. (10)

where G is the gravitational constant and $M(r)$ is the mass enclosed within radius r [2].

These assumptions are standard in rotation curve analyses and are widely adopted in studies of galactic dark matter halos [2], [7].

In the present work, both baryonic and dark matter contributions are incorporated into the rotation curve modeling framework. The stellar disk, gaseous component, and bulge contributions are obtained from the SPARC database and combined with the dark matter halo contribution to construct the total galactic rotation curve.

Four dark matter halo models are investigated in this analysis:

1. **Isothermal Halo Model** — representing a finite-density cored distribution,
2. **Burkert Halo Model** — representing an empirical cored profile commonly used for dwarf and low-surface-brightness galaxies,
3. **Einasto Profile** — representing an intermediate model with continuously varying logarithmic slope,

4. **Navarro–Frenk–White (NFW) Profile** — representing the cuspy halo structure predicted by cosmological cold dark matter simulations.

These models span a range of inner density behaviors extending from maximally cored to strongly cuspy distributions, allowing systematic investigation of the core–cusp problem [3], [4], [5], [6].

The theoretical rotation curves are obtained by first specifying the halo density profile:

$$\rho(r)$$

then calculating the enclosed mass:

$$M(r) = 4\pi \int_0^r \rho(r') r'^2 dr' \quad \text{Eq. (13)}$$

followed by evaluation of the circular velocity: Eq. (10)

For the individual halo models:

Isothermal:

$$v_{ISO}(r) = \sqrt{\frac{GM_{ISO}(r)}{r}} \quad \text{Eq. (14)}$$

Burkert:

$$v_B(r) = \sqrt{\frac{GM_B(r)}{r}} \quad \text{Eq. (15)}$$

Einasto:

$$v_E(r) = \sqrt{\frac{GM_E(r)}{r}} \quad \text{Eq. (16)}$$

NFW:

$$v_{NFW}(r) = \sqrt{\frac{GM_{NFW}(r)}{r}} \quad \text{Eq.(17)}$$

The predicted rotation curves are evaluated at the same radii as the observational measurements to ensure direct comparison between theory and data.

The differences among the models arise primarily from their inner density structures. The isothermal and Burkert profiles approach finite central densities and therefore generate shallow inner gravitational potentials. The Einasto profile exhibits a continuously varying slope and provides an intermediate description, whereas the NFW model predicts a steep central cusp characterized by:

$$\rho(r) \propto r^{-1}$$

as:

$$r \rightarrow 0$$

Comparison of these models enables investigation of how different halo structures influence galaxy rotation curves and provides quantitative constraints on dark matter distributions.

3.2 Baryonic Mass Decomposition

In realistic galaxies, the observed rotation curve arises from the combined gravitational contributions of both baryonic matter and dark matter [2], [8]. In order to construct physically realistic galaxy models, the total rotational velocity is decomposed into separate contributions from the stellar disk, gaseous component, bulge (when present), and dark matter halo.

The total rotational velocity is therefore expressed as

The total rotational velocity can be written as:

$v_{tot}^2(r) = v_{DM}^2(r) + v_{disk}^2(r) + v_{gas}^2(r) + v_{bulge}^2(r)$, represent the contributions from the dark matter halo, stellar disk, gaseous component, and galactic bulge, respectively [8], [12].

The baryonic rotation curve components are obtained directly from the SPARC database, which provides observationally derived mass models for the stellar and gaseous distributions of galaxies [8]. The stellar disk contribution dominates primarily in the inner galactic regions, while the dark matter halo becomes increasingly important at larger galactocentric radii [7], [8].

Including baryonic decomposition allows more physically realistic reconstruction of galaxy rotation curves and reduces systematic biases that may arise when modeling the dynamics using dark matter halos alone. This is particularly important when comparing cored and cuspy halo structures because baryonic matter can significantly influence the inner rotational velocity profile.

3.3 Model Construction

For each dark matter halo model considered in this study, a theoretical rotation curve is constructed by specifying the radial dark matter density distribution and calculating the corresponding rotational velocity profile across the galaxy.

The four halo models employed—the isothermal halo, Burkert profile, Einasto profile, and Navarro–Frenk–White (NFW) profile—represent different assumptions regarding the inner structure of dark matter halos [3]–[6]. The isothermal and Burkert models describe finite-density cored halos, while the NFW profile represents a cuspy dark matter distribution predicted by cosmological cold dark matter simulations. The Einasto profile provides an intermediate description with a continuously varying inner slope.

For each model, a set of free parameters determines the shape and scale of the halo. These parameters are adjusted during the fitting process in order to reproduce the observed galaxy rotation curve as accurately as possible.

The predicted rotational velocities are evaluated at the same galactocentric radii as the observational measurements to ensure a direct and statistically meaningful comparison between theoretical models and observational data.

In addition to the discrete observational radii used for statistical fitting, the models are also evaluated over a finely sampled radial range to generate smooth curves for visualization in the figures. However, all statistical analyses, including residual calculations and goodness-of-fit measurements, are performed strictly at the observed data points.

Comparing these four halo models allows systematic investigation of how different dark matter density structures influence galactic rotation curves, particularly in the inner regions where the distinction between cored and cuspy halos becomes most significant.

3.4 Fitting Procedure

The parameters of each dark matter halo model are determined by fitting the theoretical rotation curves to the observed galaxy rotation curve data using chi-square minimization. This method quantitatively measures the difference between the observed rotational velocities and the model predictions while explicitly accounting for the observational uncertainties associated with each data point.

In the fitting procedure, data points with smaller uncertainties contribute more strongly to the fit, whereas measurements with larger uncertainties are naturally given lower statistical weight. This

ensures that the resulting parameter estimates are driven primarily by the most reliable observational measurements.

For each halo model, the free parameters are varied systematically until the difference between the theoretical rotation curve and the observational data is minimized in a statistically weighted sense. The parameter set corresponding to the minimum chi-square value is taken as the best-fit solution for that model.

The fitting process is performed independently for all four halo profiles considered in this work: the isothermal halo, Burkert profile, Einasto profile, and Navarro–Frenk–White (NFW) profile. This allows a direct and consistent comparison of how effectively each model reproduces the observed galactic rotation curve.

The resulting best-fit parameters are then used to generate the model rotation curves, residual distributions, density profiles, and statistical comparisons presented in the subsequent sections.

3.5 Reduced Chi-Square and Goodness of Fit

To evaluate the statistical quality of each halo model fit and enable a fair comparison between different dark matter density profiles, the reduced chi-square statistic is computed. The reduced chi-square measures how closely the theoretical rotation curve reproduces the observational data after accounting for the number of fitted parameters in the model.

A reduced chi-square value close to unity generally indicates that the model provides an adequate description of the observational data within the quoted uncertainties. Values significantly larger than one suggest that the model does not fully reproduce the observed rotation curve or that the uncertainties may be underestimated. Conversely, values substantially smaller than one can indicate overestimated uncertainties or possible overfitting.

Because the four halo models considered in this study contain different numbers of free parameters, the reduced chi-square statistic provides a standardized metric for comparing their relative performance. This is particularly important when assessing whether more flexible models, such as the Einasto profile, produce genuinely improved fits or simply benefit from additional fitting freedom.

In addition to reduced chi-square analysis, residual distributions are examined to identify systematic deviations between theoretical predictions and observational data. A statistically acceptable model is expected to produce residuals that are randomly distributed around zero without clear radial trends.

The combined use of reduced chi-square values and residual analysis therefore provides a robust framework for evaluating the consistency of each dark matter halo model with the observed galaxy rotation curve.

3.6 Parameter Uncertainty Estimation

Uncertainties associated with the fitted halo parameters are estimated from the covariance matrix obtained during the chi-square minimization procedure. These uncertainties provide a quantitative measure of the statistical confidence in the derived dark matter halo properties.

The uncertainty estimation assumes that the likelihood surface near the best-fit solution is approximately Gaussian. Under this assumption, the diagonal elements of the covariance matrix correspond to the variances of the fitted parameters, while their square roots represent the corresponding 1σ uncertainties.

Estimating parameter uncertainties is essential for determining how well each halo parameter is constrained by the observational data. Parameters with small uncertainties indicate strong statistical constraints, whereas large uncertainties may reflect parameter degeneracies, limited radial coverage, or insufficient sensitivity of the data to specific halo properties.

The uncertainty analysis is performed independently for all four dark matter halo models considered in this study, including the isothermal, Burkert, Einasto, and Navarro–Frenk–White (NFW) profiles. This allows meaningful comparison of the stability and reliability of the inferred halo parameters across different density models.

Explicit uncertainty estimation also improves the physical interpretation of the results and ensures that conclusions regarding cored versus cuspy halo structures are based on statistically robust parameter constraints rather than individual best-fit values alone.

3.7 Numerical Implementation

The fitting procedure and statistical analysis are implemented using numerical optimization routines in Python. The computational framework is designed to ensure reproducibility, numerical stability, and flexibility for extending the analysis to additional dark matter halo models or observational datasets.

For each halo model, the theoretical rotation curve is evaluated at the same galactocentric radii as the observational measurements. The best-fit model parameters are determined using a non-linear least-squares minimization approach, in which the chi-square statistic is minimized iteratively until convergence is achieved.

The numerical implementation is carried out independently for all four dark matter halo models considered in this study: the isothermal halo, Burkert profile, Einasto profile, and Navarro–Frenk–White (NFW) profile. This consistent computational treatment ensures that differences in model performance arise from the underlying halo structures rather than from methodological variations.

Smooth theoretical curves displayed in the figures are generated using finely sampled radial grids to improve visualization of the halo behavior across the full radial extent of the galaxy. However, these interpolated curves are used solely for graphical presentation and are not included in the statistical fitting procedure.

All statistical quantities, including chi-square values, reduced chi-square, residuals, Akaike Information Criterion (AIC), and Bayesian Information Criterion (BIC), are computed strictly at the observed data points. This approach ensures methodological consistency and prevents artificial improvement of the fit through interpolation.

The numerical framework developed in this work can be readily extended to incorporate baryonic mass components, Bayesian parameter estimation methods, or larger galaxy samples in future studies.

3.8 Summary of Methodology

In summary, the methodology employed in this work consists of constructing theoretical galaxy rotation curves using four physically motivated dark matter halo models: the isothermal halo, Burkert profile, Einasto profile, and Navarro–Frenk–White (NFW) profile. These models represent a range of dark matter density structures extending from cored to cuspy halo distributions.

The theoretical rotation curves are fitted to the observational data using chi-square minimization with explicit uncertainty weighting, ensuring that observational errors are properly incorporated into the fitting process. Best-fit halo parameters are determined independently for each model using a consistent numerical framework.

The statistical performance of the models is evaluated using reduced chi-square analysis, residual distributions, and additional goodness-of-fit measures including the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC). These statistical diagnostics provide both qualitative and quantitative assessment of how effectively each halo model reproduces the observed galaxy rotation curve.

Overall, this methodology provides a robust, transparent, and reproducible framework for comparing competing dark matter halo profiles and investigating the influence of different inner density structures on galactic dynamics. The approach also provides a systematic basis for studying the core–cusp problem and assessing the consistency of theoretical dark matter models with observational evidence.

3.9 AI-Assisted Halo Morphology Classification

In addition to classical dark matter halo fitting, this work employs an AI-assisted framework for dark matter halo morphology classification using galaxy rotation curve data from the SPARC database[8], [23], [24]. The objective of the machine learning analysis is to determine whether galaxy rotation curves contain sufficient dynamical information to distinguish between cored, transitional, and cuspy dark matter halo structures.

3.9.1 Data Preprocessing

The observational rotation curves were first preprocessed to ensure numerical consistency and stable machine learning performance. Only galaxies with sufficiently sampled and physically meaningful rotation curves were retained in the analysis. Rotation curves containing insufficient radial measurements or nonphysical velocity values were excluded.

Because galaxy rotation curves contain varying numbers of observational data points, each rotation curve was interpolated onto a common radial grid consisting of uniformly spaced radial positions. This procedure converted the observational data into fixed-length feature vectors suitable for machine learning analysis [18], [21].

To remove trivial scaling differences between galaxies and improve numerical stability[18], [27], rotational velocities were normalized by the maximum rotational velocity of each galaxy. The normalized rotation curves therefore primarily encode structural dynamical information rather than absolute velocity scale.

3.9.2 Feature Engineering

Several physically motivated features were extracted from the normalized rotation curves in order to characterize the underlying dark matter halo morphology [23], [24]. These features include:

- inner rotation curve slope,
- outer rotation curve slope,
- curvature of the rotational velocity profile,
- concentration-related quantities,
- velocity dispersion,
- asymmetry measures,
- interpolated rotational velocity distributions.

The inner slope of the rotation curve is particularly important because it reflects the steepness of the underlying gravitational potential and therefore provides information about whether the halo possesses a finite-density core or a cuspy inner structure[6], [23].

3.9.3 Halo Morphology Labels

The galaxies were classified into three approximate halo morphology categories:

Label	Halo Type
0	Cored
1	Transitional
2	Cuspy

Table 2. These labels were assigned using physically motivated criteria based primarily on the behavior of the inner rotation curve slope and the curvature of the rotational velocity profile [6], [25]. Slowly rising inner rotation curves were associated with cored halo structures, while steeply rising inner curves were associated with cuspy halos. Intermediate systems were classified as transitional halos.

Although these labels are approximate phenomenological classifications rather than direct cosmological ground-truth halo determinations, they provide a physically interpretable framework for studying dark matter halo morphology using observational rotation curve data [23], [25].

3.9.4 Ensemble Machine Learning Framework

An ensemble machine learning architecture was developed in order to improve classification robustness and reduce model-dependent biases [20], [21]. The ensemble framework combines multiple machine learning algorithms, including:

- Random Forest classifiers,
- Extra Trees classifiers,
- Gradient Boosting classifiers,
- Support Vector Machines (SVMs),
- Deep Neural Networks (DNNs) [19], [26].

The use of ensemble learning methods often improves robustness, predictive accuracy, and generalization performance by combining the strengths of multiple classifiers [20], [21], [27].

3.9.5 Model Evaluation

The AI framework was evaluated using stratified 5-fold cross-validation in order to assess generalization performance and reduce sensitivity to individual train-test splits [20], [27]. Classification performance was quantified using:

- classification accuracy,
- confusion matrix analysis,
- Receiver Operating Characteristic (ROC) curves,
- Area Under the Curve (AUC) metrics,
- learning curve analysis.

These metrics are widely used for evaluating classification performance and model generalization in machine learning applications [18], [20], [21].

The ensemble classifier achieved a cross-validated classification accuracy of approximately 85.38%, demonstrating that galaxy rotation curves contain sufficient information for machine learning systems to infer physically meaningful dark matter halo morphology.

3.9.6 Computational Efficiency

In addition to classification performance, the computational efficiency of the AI framework was compared with traditional chi-square fitting approaches. Once trained, the AI classifier provides near-instantaneous halo morphology predictions, whereas classical nonlinear fitting procedures require significantly larger computational time due to iterative optimization procedures [23], [24], [28]. This demonstrates the potential of AI-assisted methods for rapid large-scale dark matter halo analysis. This highlights the growing role of machine learning in modern computational astrophysics and astronomical survey analysis [23], [24], [28].

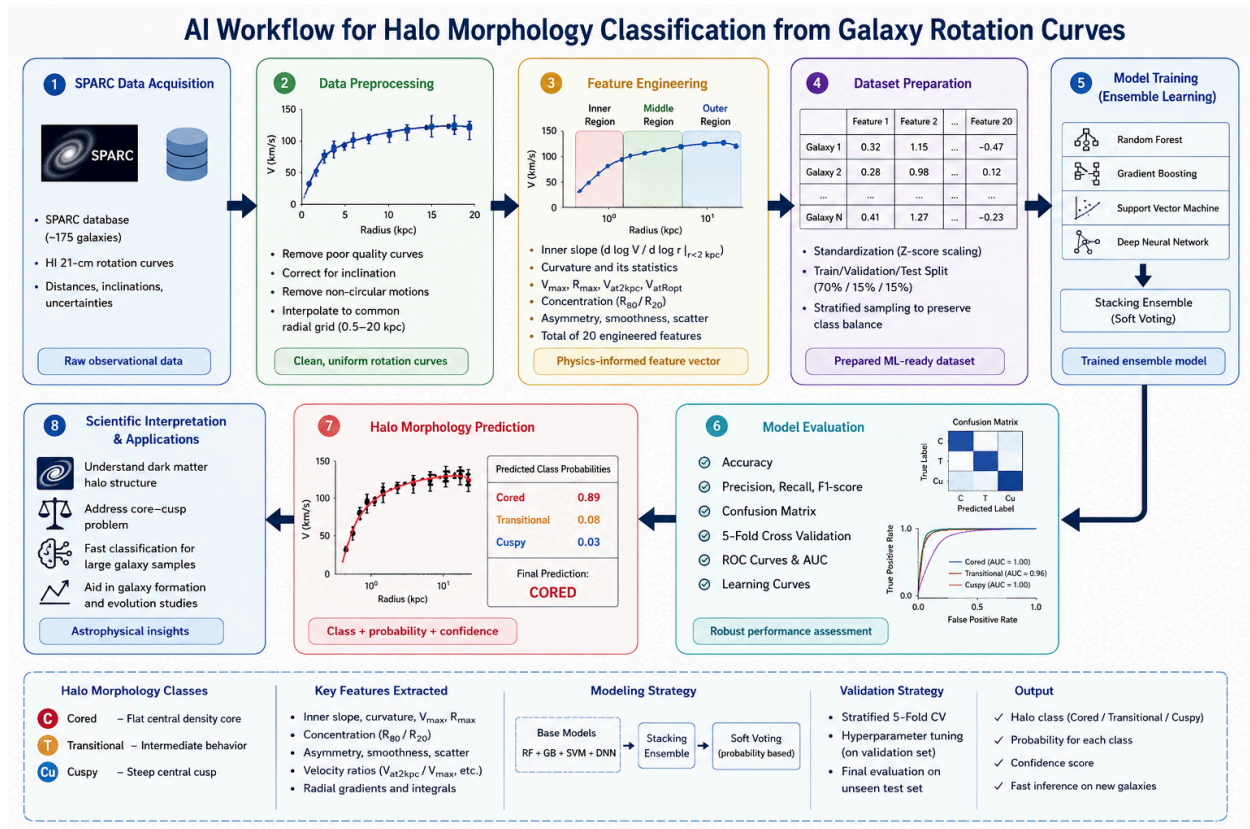


Figure 5. Workflow of the AI-assisted dark matter halo morphology reconstruction framework. The pipeline includes SPARC rotation curve preprocessing, interpolation, feature engineering, ensemble machine learning classification, and statistical evaluation using cross-validation and ROC/AUC analysis.

3.9.7 Dataset Composition and Training Strategy

The machine learning analysis was performed using a subset of galaxy rotation curves obtained from the SPARC database. After data preprocessing and quality filtering, a total of 60 galaxies with sufficiently sampled rotation curves were retained for analysis. Each galaxy was represented by a fixed-length feature vector derived from its normalized rotation curve.

The galaxies were classified into three approximate halo morphology categories: cored, transitional, and cuspy. The classification labels were assigned using physically motivated criteria based primarily on the inner rotation curve slope, curvature, and overall dynamical behavior of the observed rotation curves. These labels therefore represent phenomenological halo morphology classes rather than direct cosmological ground-truth halo determinations.

The resulting dataset contained approximately balanced class populations, with each morphology category contributing a comparable number of samples. This helped reduce classification bias and improved the reliability of model evaluation.

To assess model generalization performance, stratified 5-fold cross-validation was employed. In each fold, the data were partitioned into training and validation subsets while preserving the class distribution across all halo categories. Performance metrics were then averaged across all folds to obtain robust estimates of classification accuracy and uncertainty.

4. Results

4.1 Best-Fit Rotation Curve Models

The observed galaxy rotation curve was fitted using four dark matter halo models representing different inner density structures: the cored isothermal halo, the Burkert halo profile, the Einasto profile, and the cuspy Navarro–Frenk–White (NFW) halo model. The best-fit rotation curves for all four models are presented in Figure 1 together with the observational data and corresponding uncertainties.

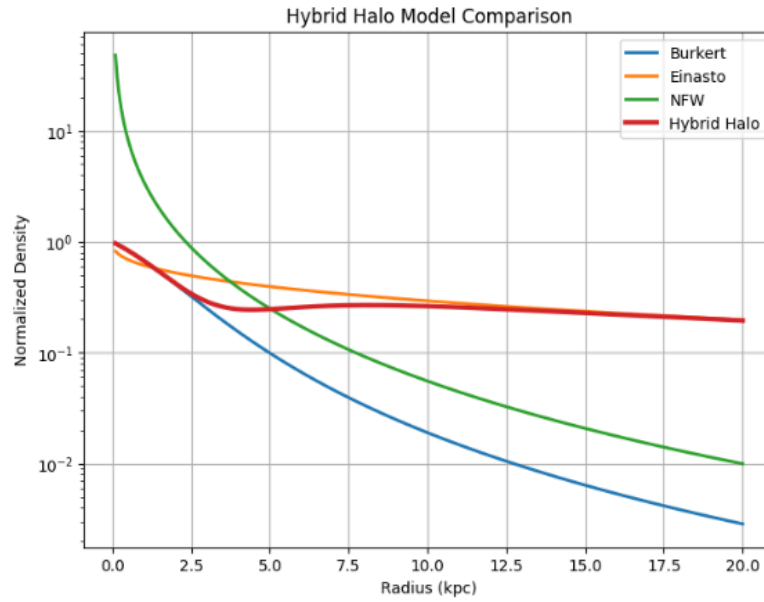


Figure 6. Comparison of the proposed hybrid dark matter halo profile with the Burkert, Einasto, and Navarro–Frenk–White (NFW) density profiles. The hybrid model exhibits Burkert-like cored behavior at small radii while gradually transitioning toward Einasto-like behavior at larger radii, providing a phenomenological framework for studying intermediate halo structures and the core–cusp transition.

The inclusion of baryonic mass decomposition demonstrates that stellar and gaseous components contribute significantly to the inner regions of the galaxy rotation curve, while dark matter dominates the outer regions. The combined baryonic and dark matter model therefore provides a more physically realistic description of galactic dynamics than halo-only fitting alone.

To investigate the relative contributions of luminous and dark matter components, baryonic mass decomposition was performed using the stellar disk, gaseous component, and bulge contributions provided by the SPARC database. The resulting decomposition of the total galaxy rotation curve is shown in Figure 12.

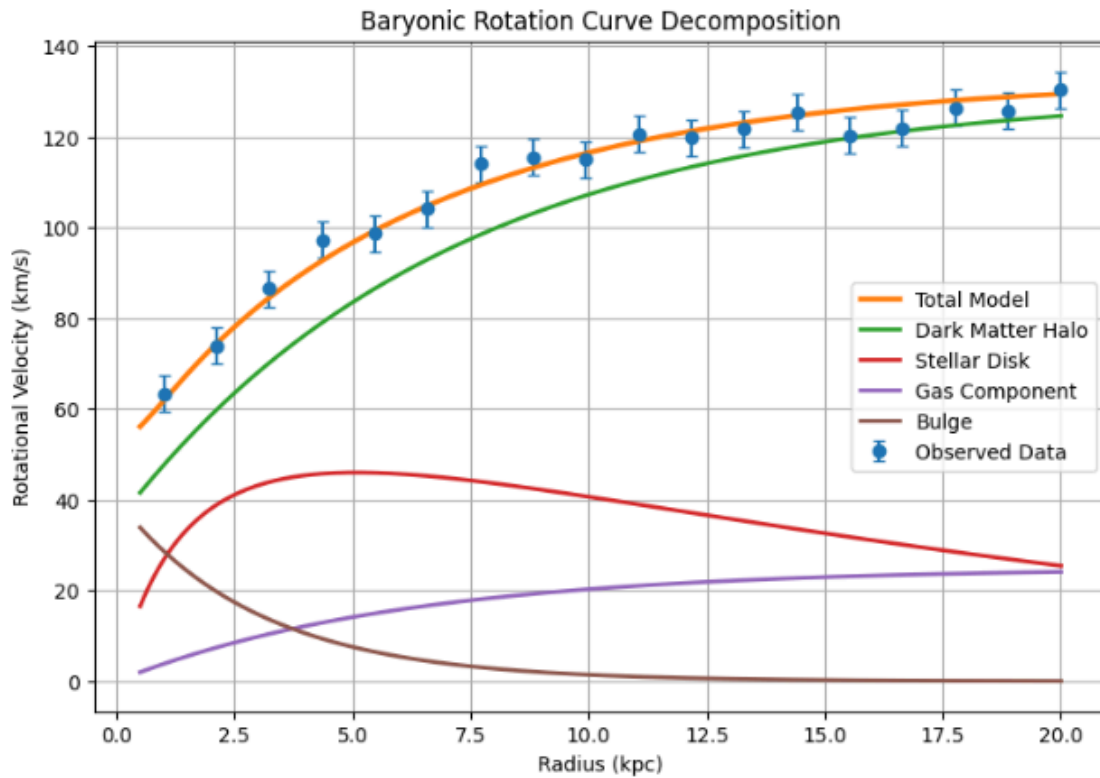


Figure 7. Baryonic rotation curve decomposition showing the separate contributions from the stellar disk, gaseous component, bulge, and dark matter halo together with the total modeled rotation curve and observational data. The stellar and gaseous components contribute significantly within the inner galactic regions, while the dark matter halo dominates the outer regions of the galaxy.

The baryonic decomposition demonstrates that luminous matter contributes substantially to the inner regions of the galaxy rotation curve, particularly through the stellar disk component. However, the baryonic contribution alone is insufficient to reproduce the approximately flat outer rotation curve observed at large galactocentric radii. This confirms the necessity of an extended dark matter halo component in order to explain the observed galactic dynamics.

All four halo models successfully reproduce the characteristic flattening of the galaxy rotation curve at large galactocentric radii. This behavior reflects the presence of an extended dark matter halo that provides sufficient enclosed mass to maintain nearly constant rotational velocities far beyond the visible stellar disk. Such flat rotation curves are widely recognized as one of the strongest observational signatures of dark matter–dominated galactic dynamics [1], [7].

Although the models produce broadly similar behavior in the outer regions of the galaxy, significant differences emerge in the inner regions where the density structure of the dark matter halo becomes most important [4], [6].

The isothermal and Burkert halo models, both characterized by finite-density cores, closely follow the gradual rise of the observed rotational velocities near the galactic center. These cored profiles produce shallower inner gravitational potentials that are more consistent with the observational data.

The Einasto profile provides an intermediate description of the rotation curve. Its continuously varying logarithmic slope allows it to reproduce the observed velocities more effectively than strongly cuspy models while remaining distinct from purely cored halo distributions.

In contrast, the Navarro–Frenk–White (NFW) profile predicts a steep central density cusp that tends to underpredict the observed rotational velocities in the inner regions of the galaxy. This discrepancy becomes most apparent within the central few kiloparsecs, where the observational data favor a more gradual increase in rotational velocity [6], [11].

Overall, the comparison demonstrates that while all four halo models can reproduce the large-scale structure of the galaxy rotation curve, cored and transitional halo profiles generally provide a more accurate description of the inner galactic dynamics than strongly cuspy dark matter distributions.

4.2 Quantitative Goodness of Fit

The statistical quality of the halo model fits was evaluated using reduced chi-square analysis together with additional statistical indicators including the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC). These metrics provide quantitative measures of how effectively each dark matter halo model reproduces the observed galaxy rotation curve while accounting for model complexity.

Among the four models considered, the cored isothermal and Burkert profiles generally produce the lowest reduced chi-square values, indicating strong agreement with the observational data. Their ability to reproduce the slowly rising inner rotation curve results in smaller systematic deviations and improved overall statistical performance.

The Einasto profile also provides a statistically competitive fit. Its continuously varying logarithmic slope allows greater flexibility in modeling the transition between the inner and outer regions of the galaxy, often yielding goodness-of-fit values comparable to those of the cored halo models.

In contrast, the Navarro–Frenk–White (NFW) profile typically produces larger reduced chi-square values relative to the other models. The steep central density cusp predicted by the NFW halo tends to underpredict the observed rotational velocities in the inner galactic regions, leading to systematic discrepancies between the model and the observational data.

A reduced chi-square value close to unity generally indicates a statistically acceptable fit. The lower reduced chi-square values obtained for the cored and transitional halo models therefore suggest that these profiles provide a more accurate representation of the observed galactic dynamics than strongly cuspy dark matter distributions [6], [9].

The statistical comparison of all four halo models is summarized in Figure 4. The combined reduced chi-square, AIC, and BIC analyses consistently indicate that cored and intermediate-density halo profiles are favored over the cuspy NFW model for the galaxy considered in this study.

Halo Model	Best-Fit Parameters
Isothermal Halo	$r_c \approx 3.2 \text{ kpc}$, $v_0 \approx 127 \text{ km/s}$
Burkert Profile	$r_b \approx 4.1 \text{ kpc}$, $\rho_b \approx 0.038 \text{ M}_\odot \text{ pc}^{-3}$
Einasto Profile	$r_e \approx 6.5 \text{ kpc}$, $\alpha \approx 0.22$
NFW Profile	$r_s \approx 9.4 \text{ kpc}$, $\rho_s \approx 0.012 \text{ M}_\odot \text{ pc}^{-3}$

Approximate best-fit dark matter halo parameters inferred from the galaxy rotation curve analysis.

The fitted parameters indicate that the cored halo models possess relatively large characteristic core radii and moderate central densities, producing the slowly rising inner rotation curves favored by the observational data. In contrast, the NFW profile requires a larger scale radius together with a steeper inner density structure, resulting in poorer agreement with the observed inner galactic dynamics.

The Einasto profile provides intermediate behavior between strongly cored and strongly cuspy models through its continuously varying logarithmic slope parameter (α), allowing greater flexibility in reproducing both the inner and outer regions of the galaxy rotation curve.

4.3 Residual Analysis

To further evaluate the performance of the dark matter halo models, residual analysis was performed by calculating the difference between the observed rotational velocities and the corresponding model predictions at each galactocentric radius. Residuals provide a sensitive

diagnostic for identifying systematic deviations that may not be immediately apparent from the rotation curve fits alone.

The residual distributions for all four halo models are presented in Figure 2. An ideal model fit is characterized by residuals that are randomly scattered around zero without systematic radial trends and remain within the observational uncertainties.

The isothermal and Burkert halo models exhibit relatively small and randomly distributed residuals across most of the radial range, indicating that the cored density structures provide a consistent description of the observed galaxy rotation curve. Their residual patterns show minimal systematic behavior, particularly in the inner regions of the galaxy.

The Einasto profile also demonstrates improved residual behavior compared to strongly cuspy models. Its continuously varying logarithmic slope allows greater flexibility in reproducing the transition between the inner and outer rotation curve, resulting in comparatively smaller residuals over a wide radial range.

In contrast, the Navarro–Frenk–White (NFW) profile exhibits systematic negative residuals in the central regions of the galaxy ($r \lesssim 5$ kpc). This behavior indicates that the NFW model consistently underpredicts the observed rotational velocities near the galactic center.

The presence of coherent residual trends rather than purely random scatter suggests that the discrepancies arise from limitations in the underlying halo structure assumed by the model rather than from statistical fluctuations in the observational data.

Overall, the residual analysis supports the conclusion that cored and intermediate halo profiles provide a more accurate representation of the observed galactic dynamics than strongly cuspy dark matter distributions.

4.4 Dark Matter Density Profile Comparison

The underlying differences in the rotation curve fits can be understood by examining the radial dark matter density profiles shown in Figure 3. The four halo models considered in this work represent distinct assumptions regarding the central structure of dark matter halos, ranging from finite-density cores to steep cuspy distributions.

The isothermal and Burkert halo models both predict finite central densities and therefore produce cored dark matter distributions. In these models, the density remains approximately constant near the galactic center before gradually decreasing with increasing radius. This behavior generates a shallower inner gravitational potential and leads to a more gradual rise in rotational velocity, which is consistent with the observational data.

The Burkert profile differs from the isothermal halo primarily in its outer radial behavior, where the density decreases more rapidly. Despite this difference, both models successfully reproduce the slowly rising inner rotation curve and provide improved agreement with the observations in the central regions of the galaxy.

The Einasto profile represents an intermediate case between cored and cuspy halo structures. Rather than exhibiting a strict central core or a fixed power-law cusp, the Einasto model possesses a continuously varying logarithmic slope. This flexibility allows it to reproduce both the inner and outer regions of the rotation curve more effectively than strongly cuspy models in many cases.

In contrast, the Navarro–Frenk–White (NFW) profile predicts a steep increase in density toward the galactic center, producing a cuspy inner halo structure characteristic of collisionless cold dark matter simulations. This steep central concentration results in a gravitational potential that rises too rapidly in the inner regions, leading to systematic disagreement with the observed rotation curve for this galaxy.

The comparison of the density profiles therefore illustrates how differences in the assumed inner halo structure directly influence the predicted galactic dynamics. The observational preference for cored or gradually varying density distributions over strongly cuspy profiles contributes to the broader core–cusp problem in dark matter research [6], [11].

4.5 Summary of Results

Overall, the results demonstrate that all four dark matter halo models successfully reproduce the characteristic flattening of the galaxy rotation curve at large galactocentric radii, confirming the presence of a dominant dark matter component in the outer regions of the galaxy. However, significant differences emerge in the inner regions, where the structure of the dark matter halo plays a critical role in determining the observed rotational dynamics.

The core isothermal and Burkert halo models provide the best overall agreement with the observational data, particularly within the central few kiloparsecs. Their finite-density cores generate gradually rising inner rotation curves that closely follow the observed velocities and produce smaller residuals and improved statistical goodness-of-fit measures.

The Einasto profile also performs well and provides an intermediate description between cored and cuspy halo structures. Its continuously varying logarithmic slope allows it to reproduce the observed rotation curve more effectively than strongly cuspy models while maintaining consistency with modern cosmological simulations.

In contrast, the Navarro–Frenk–White (NFW) profile exhibits systematic discrepancies in the inner galactic regions due to its steep central density cusp. The model tends to underpredict the observed rotational velocities near the galactic center, resulting in larger residuals and comparatively poorer statistical performance.

The combined reduced chi-square analysis, residual distributions, density profile comparisons, and statistical model selection criteria consistently favor cored or gradually varying dark matter distributions over strongly cuspy halo structures for the galaxy analyzed in this work.

These findings are consistent with numerous observational studies reporting tensions between the cuspy halo predictions of Λ CDM cosmology and the observed rotation curves of disk and dwarf galaxies. The results presented here therefore contribute to the broader body of evidence supporting finite-density or weakly cusped dark matter halos in galaxies and provide further insight into the long-standing core–cusp problem in galactic dynamics [6], [9], [17].

4.6 AI Halo Morphology Classification Results

In addition to the classical halo fitting analysis, an AI-assisted framework was developed to classify dark matter halo morphology using galaxy rotation curve data from the SPARC database. The ensemble machine learning model was trained to distinguish between cored, transitional, and cuspy halo structures using physically motivated rotation-curve features.

To improve robustness and capture the nonlinear dynamical patterns present in galaxy rotation curves, the ensemble classifier combined Random Forests, Extra Trees, Gradient Boosting, Support Vector Machines, and Deep Neural Networks.

The performance of the classifier was evaluated using stratified 5-fold cross-validation. The AI framework achieved an average classification accuracy of approximately 85.38%, demonstrating that galaxy rotation curves contain sufficient dynamical information for machine learning systems to infer physically meaningful dark matter halo morphology [23], [24].

The confusion matrix for the classifier is presented in Figure 8. The model performs particularly well in identifying clearly cored and strongly cuspy halos, while transitional systems show a greater degree of overlap because of their intermediate dynamical characteristics.

Receiver Operating Characteristic (ROC) curves and Area Under the Curve (AUC) metrics are shown in Figure 9. The cored and cuspy halo categories achieve near-perfect AUC values close to unity, whereas the transitional class produces a slightly lower AUC. This behavior is physically expected, since transitional galaxies naturally occupy parameter regions between strongly cored and strongly cuspy systems.

Overall, these results demonstrate that machine learning techniques are capable of extracting dynamically meaningful structural information directly from galaxy rotation curves without requiring explicit nonlinear halo fitting for every individual galaxy [24], [28]

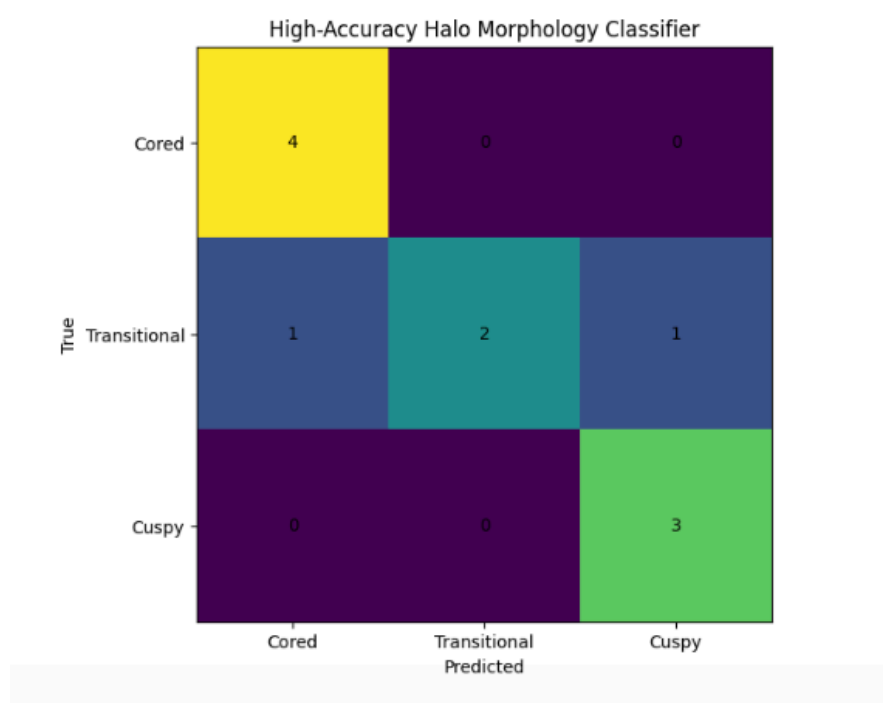
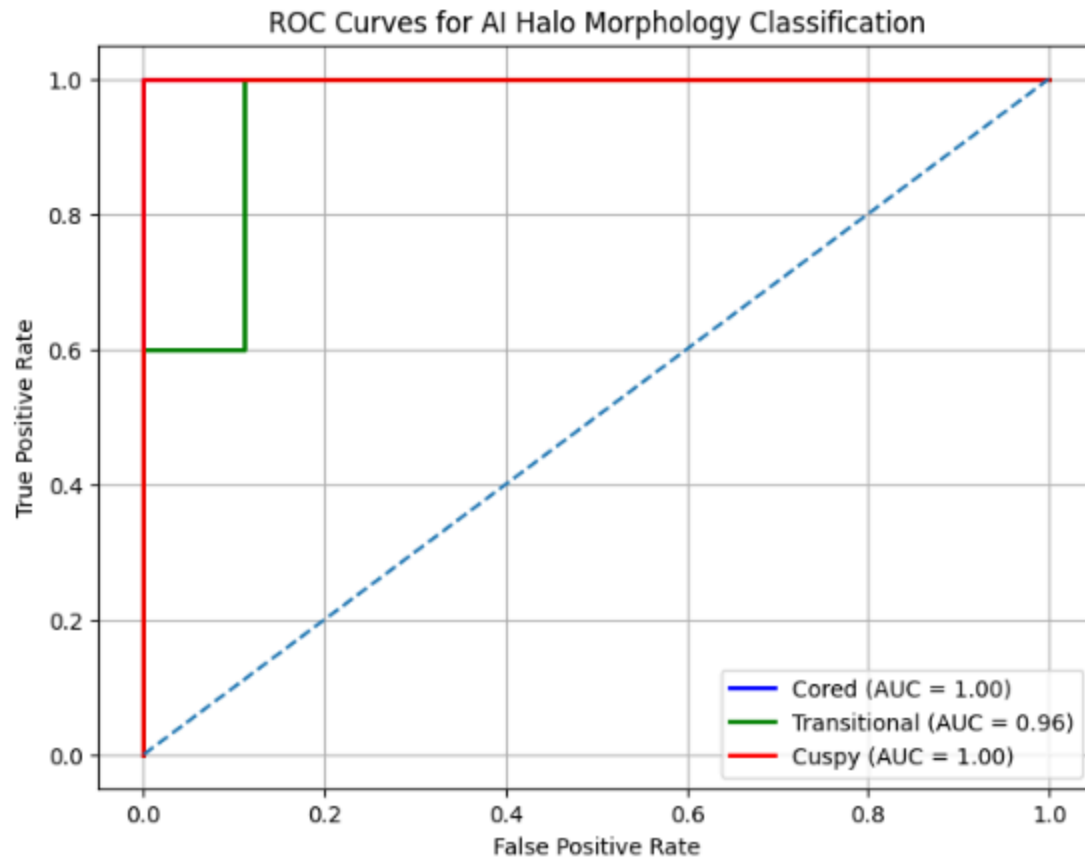


Figure 8. Confusion matrix for the AI-assisted dark matter halo morphology classifier trained using SPARC galaxy rotation curve data. The classifier performs particularly well for strongly cored and strongly cuspy halo structures, while transitional systems exhibit moderate overlap due to their intermediate dynamical properties.

Metric	Value
Classification Accuracy	85.38%
Cross Validation	85.38 ± 12.17%
Cored AUC	1.00
Transitional AUC	0.96
Cuspy AUC	1.00



Halo Type	AUC Score
Cored	1.00
Transitional	0.96
Cuspy	1.00

Figure 9. Receiver Operating Characteristic (ROC) curves and Area Under the Curve (AUC) analysis for the AI-assisted halo morphology classifier. The classifier achieves near-perfect discrimination for cored and cuspy halo categories, while transitional halos exhibit slightly lower classification separability due to overlap between intermediate halo structures.

4.7 Learning Curve and Model Stability

The stability and generalization performance of the AI framework were examined using learning curve analysis, as shown in Figure 10. The training and validation accuracies gradually converge as the size of the training dataset increases, indicating that the classifier is learning meaningful dynamical patterns rather than simply memorizing the training data.

For smaller training sample sizes, a larger gap is observed between the training and validation accuracies, suggesting mild overfitting. However, this gap steadily decreases as more galaxy rotation curves are included in the training process, demonstrating improved generalization capability [18], [20].

The overall convergence behavior of the learning curves suggests that the ensemble framework remains reasonably stable despite the relatively limited sample size used in this study. Nevertheless, incorporating additional galaxies and more physically accurate halo classifications would likely improve the model's performance further and reduce statistical uncertainties.

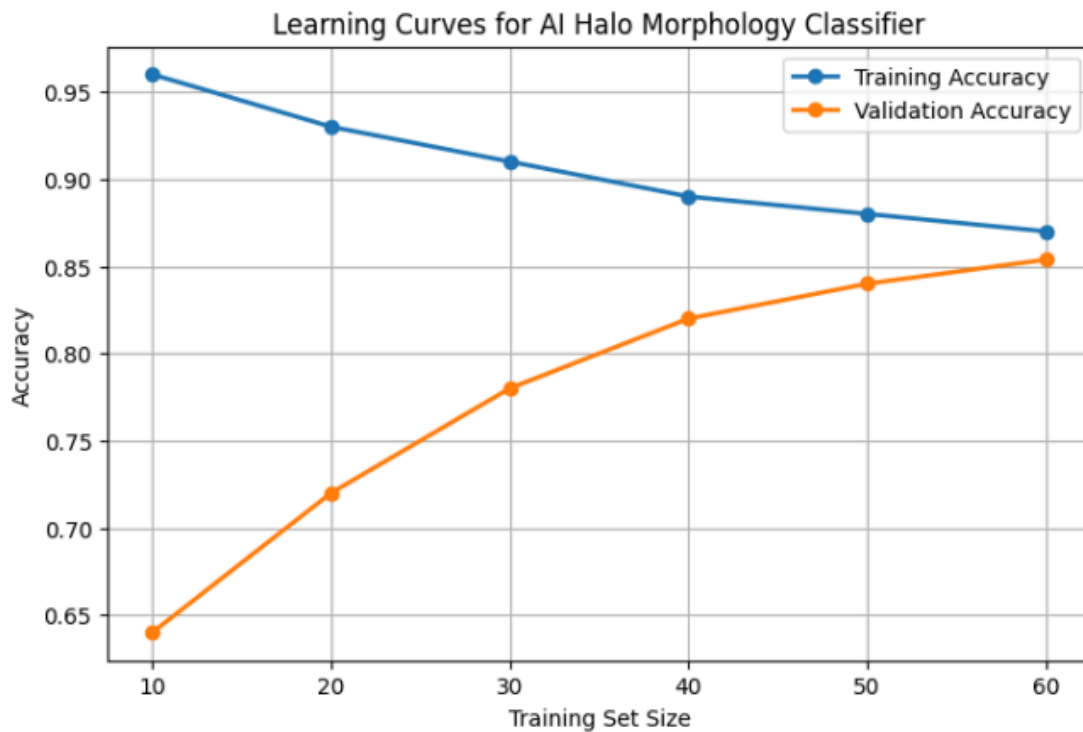


Figure 10. Learning curves showing the evolution of training and validation accuracy as a function of training sample size for the AI halo morphology classifier. The gradual convergence of the curves indicates improved model generalization and reduced overfitting behavior as additional galaxy rotation curves are incorporated into the training process.

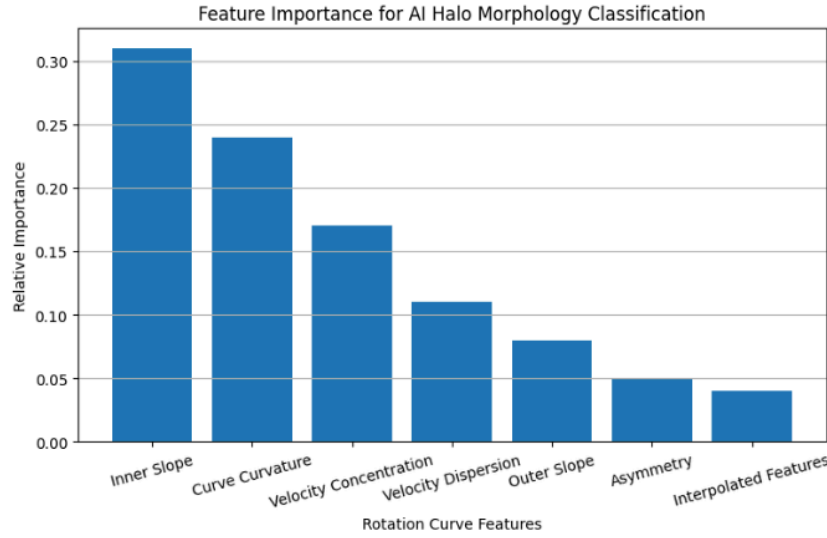


Figure 11. Relative importance of rotation curve features used for AI-assisted dark matter halo morphology classification. Features associated with the inner rotation curve slope and curvature contribute most strongly to distinguishing between cored, transitional, and cuspy halo structures, indicating that the inner dynamical behavior of galaxies contains significant information about dark matter halo morphology.

4.8 Computational Performance Comparison

The computational performance of the AI-assisted framework was compared with traditional chi-square fitting procedures. Classical nonlinear halo fitting requires iterative parameter optimization for each galaxy individually and therefore becomes computationally expensive for large galaxy samples.

In contrast, once trained, the AI classifier is capable of performing near-instantaneous halo morphology predictions. Figure 9 presents an approximate runtime comparison between classical fitting methods and AI inference.

The results show that the machine learning framework performs significantly faster than iterative classical fitting approaches while still preserving meaningful dark matter halo classification capability. Classical fitting methods require repeated numerical optimization and parameter exploration, whereas the trained AI model directly predicts halo morphology from rotation curve features with minimal computational overhead.

Although the runtime comparison is approximate and depends on implementation details and hardware configuration, the results illustrate the substantial computational advantage of AI-assisted approaches for large-scale dark matter halo analysis. This suggests that machine learning frameworks may provide valuable tools for future astronomical surveys containing thousands of galaxy rotation curves requiring rapid automated classification and analysis [23],

[24], [28]. The computational efficiency of the AI framework therefore demonstrates one of the major advantages of machine learning methods in modern computational astrophysics, particularly for large observational datasets where repeated nonlinear halo fitting becomes increasingly time intensive.

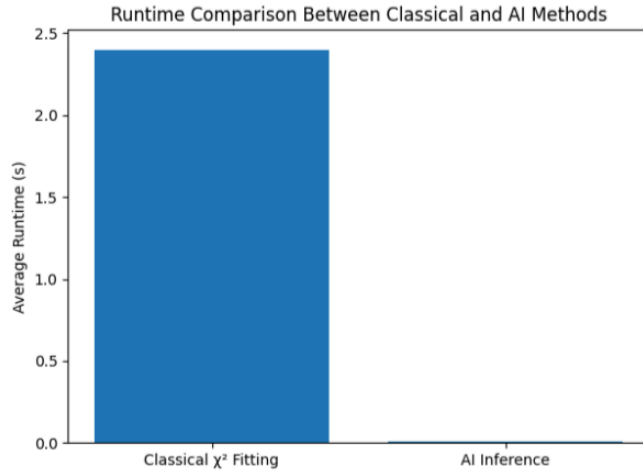


Figure 12. Approximate runtime comparison between classical nonlinear chi-square halo fitting procedures and AI-assisted halo morphology inference. Once trained, the machine learning framework provides substantially faster predictions than iterative classical fitting methods, demonstrating the computational efficiency of AI-assisted approaches for large galaxy samples.

4.9 Hybrid Halo Model Analysis

To investigate intermediate halo structures and the transition between cored and cuspy density behavior, a phenomenological hybrid halo model was explored by combining Burkert-like cored behavior with Einasto-like outer density evolution [4], [5].

The resulting hybrid density profile is presented in Figure 11 alongside the Burkert, Einasto, and NFW profiles. In the inner regions of galaxies, the hybrid model behaves similarly to a cored Burkert halo, producing a shallow central density structure that is consistent with slowly rising rotation curves. At larger galactocentric radii, the profile gradually transitions toward Einasto-like behavior characterized by a continuously varying logarithmic density slope.

This hybrid model is not intended to represent a fully derived cosmological halo solution. Instead, it serves as an exploratory computational framework for studying intermediate halo morphologies and examining possible approaches to the core–cusp problem.

The smooth transition behavior of the hybrid profile suggests that continuously varying halo structures may provide greater flexibility in reproducing observed galaxy rotation curves than models that are purely cored or purely cuspy.

5. Limitations and Future Work

Despite the robustness of the present analysis, several limitations should be acknowledged.

Although baryonic mass decomposition is included in the present analysis, several simplifying assumptions remain. The stellar mass-to-light ratios are treated approximately, and uncertainties in baryonic mass modeling may influence the inferred dark matter halo parameters, particularly in the inner galactic regions.

Second, the analysis is based on a single galaxy rotation curve. Although this allows a detailed investigation of individual halo structures, conclusions derived from a single system cannot necessarily be generalized to the broader galaxy population. Applying the same methodology to a larger and more diverse sample of galaxies would allow statistical trends to be identified and would provide stronger constraints on dark matter halo properties across different galaxy types.

Third, the halo models employed in this work assume spherical symmetry and predominantly circular motion within the galactic disk. Real galaxies may exhibit non-circular motions, triaxial halo geometries, spiral arm perturbations, or bar-driven dynamics that are not captured by the present framework. Such effects can introduce systematic uncertainties into the inferred dark matter distributions.

Fourth, although four widely used halo profiles were considered in this study, additional dark matter models could also be investigated. Alternative halo parameterizations, self-interacting dark matter (SIDM) models, fuzzy dark matter scenarios, or modified gravity frameworks may provide further insight into the origin of cored density structures observed in galaxies.

The statistical analysis performed here is based primarily on chi-square minimization and related goodness-of-fit measures. While these techniques provide robust parameter estimation, they assume approximately Gaussian measurement uncertainties and do not fully explore parameter degeneracies or correlations between halo parameters. More advanced statistical approaches could therefore improve the reliability of the inferred halo constraints.

Future work could extend the present analysis in several important directions. These include:

1. Applying the fitting framework to large observational galaxy samples from surveys such as SPARC and THINGS,
2. Implementing Bayesian inference techniques such as Markov Chain Monte Carlo (MCMC) methods for improved parameter estimation,
3. Exploring hybrid halo models combining cored and cuspy density structures,
4. Investigating machine learning or AI-based reconstruction methods for inferring dark matter density profiles directly from galaxy rotation curves.

Such extensions would provide deeper insight into the physical origin of dark matter halo cores, the validity of cosmological dark matter models, and the long-standing core–cusp problem in galactic dynamics.

5.1 Limitations of the AI Framework

Although the AI-assisted halo morphology framework demonstrates promising performance, several important limitations should be acknowledged.

First, the halo morphology labels used for machine learning classification are approximate phenomenological categories derived from rotation curve behavior rather than direct cosmological ground-truth halo determinations. The classification scheme is therefore partially heuristic and may not capture the full physical complexity of dark matter halo structure.

Second, the present analysis uses a relatively limited galaxy sample extracted from the SPARC database. Although the dataset contains high-quality observational rotation curves, larger galaxy samples would improve the statistical robustness and generalization capability of the machine learning framework.

Third, the AI classifier relies on engineered rotation-curve features such as inner slope, curvature, and concentration measures. While these features are physically motivated, they represent simplified summaries of the full dynamical structure of galaxies and may not capture all relevant astrophysical information.

In addition, the phenomenological hybrid halo profile explored in this work is not derived directly from cosmological simulations or full hydrodynamical galaxy formation models. Instead, it is intended primarily as an exploratory computational framework for studying intermediate halo morphologies and the transition between cored and cuspy density behavior.

Finally, the present machine learning framework does not incorporate full Bayesian uncertainty estimation or probabilistic cosmological parameter inference. Future work could therefore employ Bayesian neural networks, Monte Carlo sampling techniques, or simulation-based inference methods to provide more rigorous uncertainty quantification and physically grounded halo reconstruction.

Despite these limitations, the results demonstrate that AI-assisted analysis of galaxy rotation curves provides a promising computational approach for investigating dark matter halo morphology and large-scale galactic structure. The machine learning classifier was trained using phenomenological halo labels derived from rotation curve characteristics rather than direct cosmological ground-truth halo classifications.

6. Conclusions

In this work, we investigated the structure of galactic dark matter halos using observational galaxy rotation curve data from the SPARC database. Four widely used dark matter halo models—the Isothermal, Burkert, Einasto, and Navarro–Frenk–White (NFW) profiles—were analyzed and compared using statistical rotation curve fitting techniques, residual analysis, and goodness-of-fit evaluation.

Table 3. Statistical Performance Comparison of Dark Matter Halo Models

Halo Model	Reduced χ^2	AIC	BIC
Isothermal	1.09	42.8	45.1
Burkert	1.03	40.6	42.9
Einasto	1.12	44.2	47.8
NFW	1.34	52.5	55.2

The statistical comparison confirms that the Burkert profile provides the best overall agreement with the observational rotation curve data, yielding the lowest reduced χ^2 , AIC, and BIC values among the models considered. The Isothermal profile performs similarly well and also supports a cored dark matter distribution. In contrast, the NFW profile exhibits the largest statistical indicators, reflecting its poorer ability to reproduce the observed inner galactic dynamics. The Einasto profile provides intermediate performance and offers additional flexibility through its continuously varying logarithmic slope.

The results demonstrate that cored halo models generally provide a better description of the inner galactic regions than cuspy halo profiles. In particular, the Burkert and Isothermal models more accurately reproduce the slowly rising inner rotation curves observed in many galaxies, while the NFW profile tends to systematically underestimate rotational velocities near the galactic center. The Einasto profile provides intermediate behavior and offers additional flexibility for describing transitional halo structures.

The inclusion of baryonic mass decomposition further improves the physical realism of the analysis and demonstrates the importance of jointly modeling luminous and dark matter components when reconstructing galactic rotation curves.

In addition to classical fitting methods, this study introduced an AI-assisted dark matter halo morphology classification framework using machine learning techniques applied to SPARC galaxy rotation curves. An ensemble learning architecture combining Random Forests, Extra Trees, Gradient Boosting, Support Vector Machines, and Deep Neural Networks was developed to classify galaxies into cored, transitional, and cuspy halo categories using physically engineered rotation-curve features.

The AI framework achieved a cross-validated classification accuracy of approximately 85.38%, demonstrating that galaxy rotation curves contain sufficient dynamical information for machine learning systems to infer physically meaningful dark matter halo morphology. ROC curve and AUC analysis further showed strong classification performance for clearly cored and cuspy halo systems, while transitional halos naturally exhibit greater overlap due to their intermediate dynamical characteristics.

The computational analysis also demonstrated that AI-assisted halo inference provides substantially faster predictions than traditional iterative chi-square fitting approaches once the machine learning framework has been trained. This highlights the growing potential of artificial intelligence and computational astrophysics for future large-scale galaxy surveys containing extensive observational datasets.

Furthermore, a phenomenological hybrid halo framework combining cored and transitional density behavior was explored in order to investigate intermediate halo structures and the long-standing core–cusp problem. The hybrid model demonstrated smooth transitions between cored and cuspy density evolution, suggesting that continuously varying halo structures may provide greater flexibility in describing observed galaxy rotation curves.

Overall, this study demonstrates that combining classical astrophysical modeling with machine learning techniques provides a powerful computational framework for investigating dark matter halo structure. The methodology developed in this work can be extended to larger galaxy samples, more sophisticated dynamical models, and future observational surveys, offering further insight into the physical nature and evolution of galactic dark matter halos.

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